ASIRIVE // CORTEX

ASIRIVE CORTEX // HYBRID ARCHITECTURE

Navigation INDEPENDENCE

ASIRIVE CORTEX

A **$186 BOM ($250 retail)** screenless, audio-first navigator co-designed with **SAVH** to replace $2,500+ legacy devices



IRFAN
Hardware Lead

The 5 Specialists

Guardian Edge Layer (<100ms): Raspberry Pi 5 + Hailo-8L NPU running offline YOLO models to **intercept cane blind spots** (head-height obstacles, steps, fast approaches).

Gemini Live Brain: Multimodal cloud orchestrator managing dynamic tool calling for live **LTA DataMall bus** arrivals, Google Maps routing, and **5-mode auto-switching.**

Open-Ear Audio & Fail-Safe: Cartesia TTS spatial feedback keeping ambient hearing unblocked, backed by a 3s **connectivity monitor for zero silent failures.**

https://www.asirive.com/cortex